MENG 4010 - INDUSTRIAL NETWORKING

VLAN SEGMENTATION USING SCALANCE XB208

Student Name & Number: .....

Lab Date:.....

Objective

In this lab, students will configure Virtual Local Area Networks (VLANs) using a SCALANCE XB208 managed industrial switch. VLAN segmentation is used to logically separate traffic within an industrial Ethernet network.

Upon successful completion of this lab, students will be able to:

- Understand and explain the function of VLANs in industrial networking
- Configure 802.1Q VLANs via the Web-Based Management (WBM) interface of the SCALANCE XB208
- Create VLAN segments within a single switch and extend VLANs across two switches using trunk ports.
- Test VLAN isolation and trunking using basic tools such as ping

Background

VLANs allow a single physical network infrastructure to support multiple logical networks, improving security, performance, and manageability. In an industrial context, VLANs can separate automation traffic from administrative or guest traffic.

The SCALANCE XB208 switch supports IEEE 802.1Q VLANs, and its Web-Based Management (WBM) interface allows users to configure VLANs using a graphical user interface.

There are two main scenarios for VLAN implementation in this lab:

- Scenario 1: Single Switch VLAN Segmentation – Ports are assigned to different VLANs to restrict communication across them.
- Scenario 2: Multi-Switch VLAN Trunking – VLAN configuration is extended across two XB208 switches using trunk ports that retain VLAN tags.

Demonstration Requirements

To successfully complete the lab, students must demonstrate the following:

Scenario 1: VLAN Configuration

- Enable VLAN mode on the SCALANCE XB208 via the WBM interface
- Create two VLANs (VLAN 1 for admin including host computer, VLAN 2 for field devices)
- Assign specific ports to VLANs based on lab topology

Network Behavior Validation

- Use ping from a PC in VLAN1 to test access to:
  - SCALANCE switch web interface (should succeed)
  - Devices in VLAN2 (should fail)
- Use ping from a PC in VLAN2 to test access to:
  - SCALANCE switch web interface (should fail)
  - Devices in VLAN2 (should succeed)

Optional Extension Task (Bonus)

To further enhance your understanding of VLAN segmentation and multi-switch communication using SCALANCE switches, complete the following extended configuration task:

Scenario 2: VLANs Across Two Switches

Expand your VLAN configuration to span two SCALANCE XB208 switches connected via trunk ports.

Instructions:

1. **Topology Setup**:
   - Connect port P0.8 on Switch A to port P0.8 on Switch B.
   - Assign ports P0.2 and P0.4 on both switches to VLAN 2 (untagged).
   - Configure P0.8 on both switches as a trunk port using "Switch-Port VLAN Trunk" mode.
2. **IP Configuration**:
   - Ensure each end device in VLAN 2 a unique IP in the same subnet
   - Ensure the devices can be reached via ping.
3. **Verification**:
   - Verify that devices in VLAN 2 on different switches can communicate through the trunk port.
   - Confirm that a device in VLAN 1 cannot access VLAN 2 devices.
   - Document the success/failure of each ping test.

Lab Report Requirements

The lab report must contain:

- Instructor's Confirmation

The course instructor's signature is required to verify the completion of the lab. Please ensure this is included with your report's submission.

- Assignment Summary
  Provide a clear summary of the assignment, detailing the objectives, a description of the work completed, and your conclusions.
- Configuration Evidence
  - Screenshots of VLAN tables, trunk port settings.
  - Ping test results from both VLANs
  - Port assignment diagrams.
- Deliverables for Optional Extension Task
  - VLAN configuration screenshots from both switches.
  - A labeled topology diagram.
  - Ping test results showing isolation and connectivity.
  - Short paragraph explaining why trunking is necessary for cross-switch VLAN operation.
- The lab report will be assessed based on the lab marking rubric provided on Blackboard. Please review the rubric to ensure all criteria are met.

Application Photos

	